

Milestone project -1

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Cohort -12

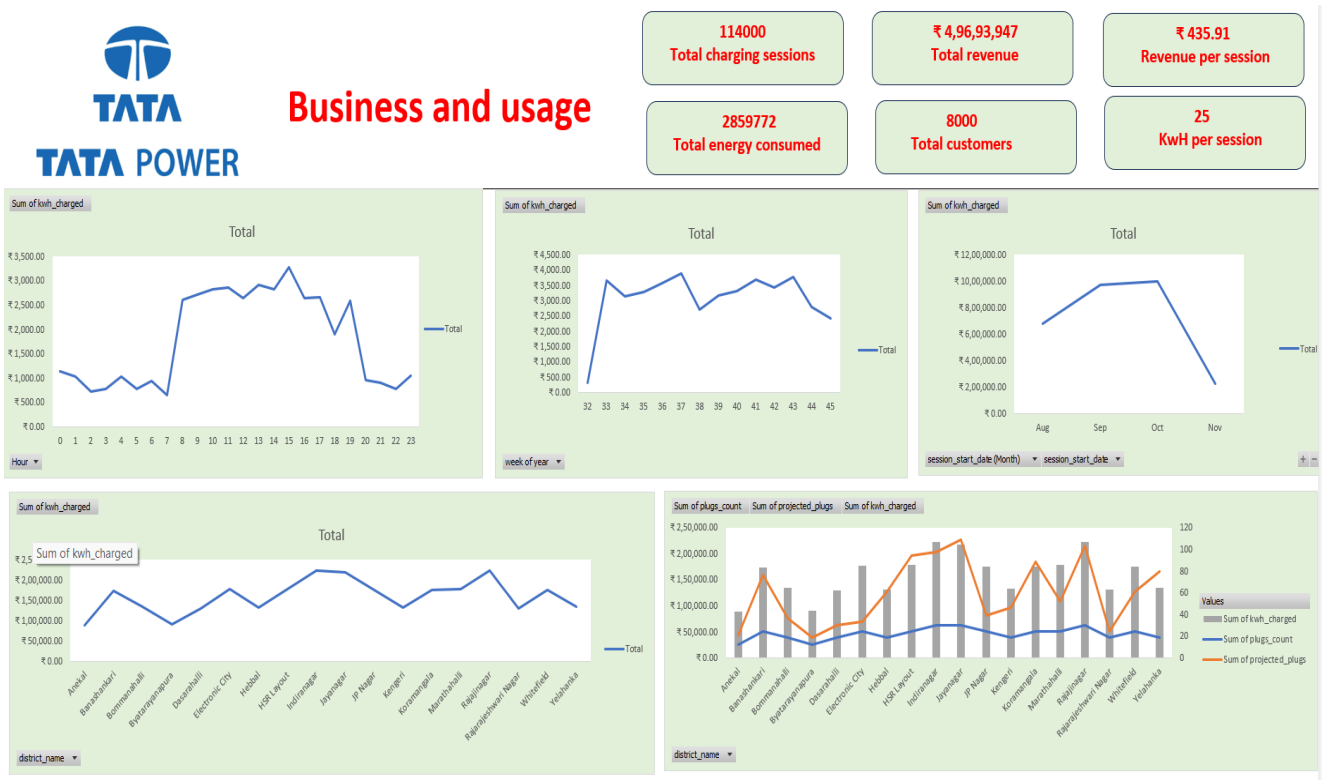
Business and Usage Overview

Problems to address:

- How is charging demand evolving over time
- Are stations able to handle growing demand
- What is the charging behaviour pattern by time of day

KPIs to track:

- Total charging sessions
- Total energy consumed (kWh charged)
- Total revenue (from charging sessions)
- Number of active customers
- Average revenue per session
- Average kWh per session



Graph 1: Energy Usage by Hour (0–23 hrs)

- Low charging activity during midnight to early morning (0–7 hrs).
- Sharp increase after 8 AM, indicating office commute charging.
- Peak usage between 10 AM – 4 PM.
- Drop after 8 PM

Interpretation:

Customers mostly charge during working hours → charging stations near offices/commercial areas are heavily utilized.

Graph 2: Energy Usage by Week of Year

- Demand is relatively stable.
- Peaks around **week 37–43**.
- Slight decline after week 44.

Interpretation

Possible seasonal/business cycle impact.
Could indicate festive period EV travel increase followed by normalization.

Graph 3: Monthly Revenue Trend (Aug–Nov)

- Revenue rises from **August to October**
- Sharp drop in **November**

Possible Reasons

- Reduced charging sessions
- Seasonal fluctuations
- Operational downtime / station maintenance.

Graph 4: District-wise Energy Consumption

High Demand Districts

- Jayanagar
- Rajajinagar

- HSR Layout

Low Demand Districts

- Anekal
- Byatarayanapura

Interpretation:

Urban premium locations generate higher charging traffic.

Expansion should prioritize top-performing districts.

Graph 5: Existing Plugs vs Projected Plugs vs Energy Usage

It shows Infrastructure adequacy.

- **Grey bars:** Existing plugs
- **Orange line:** Projected plugs needed
- **Blue line:** Current usage

Insights

- Districts like **Jayanagar / Rajajinagar** show projected demand > current capacity.

Interpretation

Infrastructure gap exists.
Future charger deployment should focus here.

Station Utilization and Operational Efficiency

Problems to address:

- Which stations are overutilized and may require capacity expansion
- Which stations are underutilized and need operational improvement
- Are available plugs being efficiently used across locations
- What is the load distribution across districts and stations

KPIs to track:

- Number of sessions per station
- Average utilization per station (sessions per plug)
- Plugs count vs actual usage
- Average kWh per station
- District wise station density
- Sessions per plug ratio



Graph 1: Projected Plug Utilization % by District

Highest utilization:

- Yelahanka
- HSR Layout
- Koramangala

Lowest utilization:

- Dasarahalli
- Electronic City

Interpretation:

Some districts are near capacity saturation.
Expansion priority required.

Graph 2: kWh Charged vs Plug Count

It shows Operational load relative to infrastructure.

- Some districts have high plug count but average usage → underutilization.
- Others show strong charging despite fewer plugs → efficient deployment.

Interpretation

Infrastructure optimization opportunity.

Graph 3: Session Count vs Energy Delivered by Station ID

It shows Individual station performance.

- Most stations perform consistently.
- A few stations have lower sessions → possible poor location or downtime.

Action

Investigate low-performing station IDs for relocation or maintenance.

Customer and Demand Behavior

Problems to address:

- What type of customers contribute most to usage and revenue
- How does income tier influence charging behavior
- What are the usage patterns across different EV models and battery capacities
- Are there segments with high potential but low engagement

KPIs to track:

- Unique customers count
- Sessions per customer
- Average kWh per customer
- Revenue per customer
- Usage by income tier
- Usage by car model and battery capacity



Graph 1- Customer Count vs Energy Usage by Income Tier

- Low-Mid income segment has highest customer count.
- Mid-range customers contribute highest total charging cost.

Interpretation

Middle-class EV owners are the strongest charging customer base

Graph 2- Sessions vs Average Cost vs Average kWh

- **Low-Mid tier** shows highest sessions.
- High income customers charge less frequently but higher ticket value.

Interpretation

Different pricing plans could improve monetization.

Graph 3- Car Model vs Battery Capacity vs Charging Usage

Higher battery EVs tend to consume more charging energy.

Popular heavy-usage EVs:

- Tata Nexon EV
- MG ZS EV
- Hyundai Kona

Interpretation

Large-battery EV owners are premium charging customers.

Graph 4 - Income Tier vs Session Volume

Session count increases toward mid-range income group.

Interpretation

Mass-market EV adoption is driving demand more than luxury EV segment.

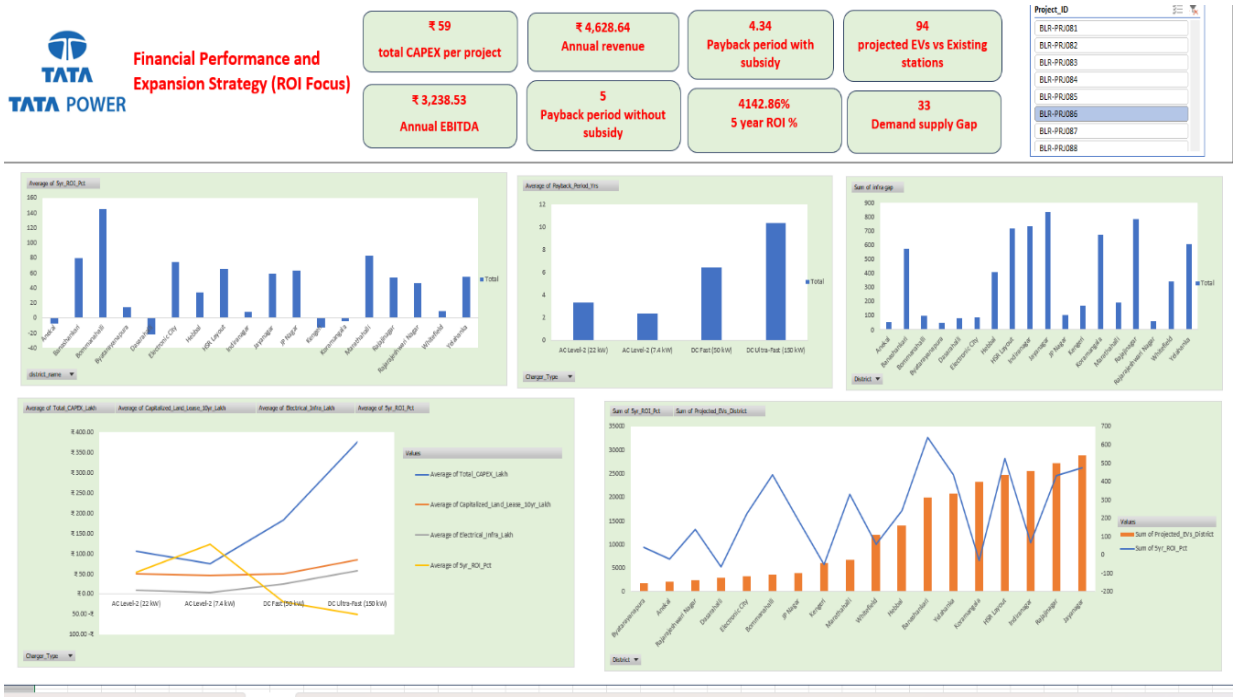
Financial Performance & Expansion Strategy

Problems to address:

- Which districts and projects deliver the highest ROI
- What is the payback period for different charging setups
- How do costs such as CAPEX, land lease, and infrastructure impact profitability
- Which locations should be prioritized for new station deployment based on demand and ROI
- What is the gap between existing infrastructure and projected EV demand

KPIs to track:

- Total CAPEX per project
- Annual revenue and EBITDA
- Payback period (with and without subsidy)
- 5 year ROI percentage
- Existing stations vs projected EVs
- Demand supply gap (projected EVs vs projected plugs)



Graph 1 : District-wise 5-Year ROI %

Highest ROI

- Bommanahalli
- Mahalakshmi
- Electronic City

Lowest ROI

- Anekal

Interpretation

Expansion should prioritize high-ROI urban zones.

Graph 2 : Payback Period by Charger Type

Fastest Payback

- AC Level-2

Slowest

- DC Ultra-Fast (150kW)

Interpretation

AC chargers are safer financially.
Ultra-fast chargers should be deployed only in premium/high-demand locations.

Graph 3: Infrastructure Gap by District

High Gap

- Jayanagar
- HSR Layout
- Rajajinagar

Interpretation

Immediate deployment opportunity.

Graph 4: CAPEX Components by Charger Type

It Shows:

- Total CAPEX

- Land lease cost
- Electrical infra cost
- ROI %

Insight

Ultra-fast chargers have highest setup cost but can justify premium pricing.

Graph 5 : Projected EV Growth vs ROI

Districts with both:

- High projected EV growth
- High ROI

are ideal expansion targets.

Best strategic districts:

- Jayanagar
- Bommanahalli
- HSR Layout
- Rajajinagar

Conclusions

This dashboard overall shows **Tata Power EV network is profitable, scalable, and has strong urban demand with clear expansion opportunities.**